

Abstract

This review synthetically bridges case theory, categorial grammar, enriched category theory, and active inference into a single algebraic framework centered on the *cognitive case diagram*. By formalizing case systems as categories (objects mapping to case roles and morphisms to grammatical relations) and alignment typologies as functors, we render sentence structure as compact-closed string diagrams computable via the DisCoCat framework. These diagrams are not merely descriptive notation; drawing upon the cognitive science of diagrammatic reasoning and the embodied constraint of sketching thought onto a two-dimensional plane, they are executable proofs of grammaticality. We review the historical foundations from Pāṇini's *kāraṇa* theory to Fillmore's deep semantic cases and Mel'čuk's relational networks, tracing how morphosyntactic inflection evolved into the structural skeletons of generative models where comprehension reduces to free-energy minimization. We demonstrate that this topological approach constitutes the formal algebra underlying the distributional meaning representations of